

Math 81: Abstract Algebra

Prishita Dharampal

Problem 1. Let F be a field and K/F and L/F be subextensions of a field extension M/F .

- (a) Prove that if $K = F(\alpha_1, \dots, \alpha_n)$ and $L = F(\beta_1, \dots, \beta_m)$ are finitely generated, then

$$KL = F(\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_m).$$

- (b) Prove that if K/F and L/F are finite then KL/F is finite and

$$[KL : F] \leq [K : F][L : F],$$

with equality if and only if K/F and L/F are linearly disjoint.

Hint. Prove that if $x_1, \dots, x_n$ is an F -basis for K and $y_1, \dots, y_m$ is an F -basis for L , then the products $x_i y_j$ for $1 \leq i \leq n$ and $1 \leq j \leq m$ span KL/F and are an F -basis if and only if K/F and L/F are linearly disjoint.

- (c) Prove that finite extensions K/F and L/F of relatively prime degree are linearly disjoint.
- (d) Prove that if K/F and L/F are linearly disjoint then $K \cap L = F$. Find an example showing that the converse is false.
- (e) Prove that if K/F and L/F are linearly disjoint finite Galois extensions then KL/F is Galois and the map

$$\text{Gal}(KL/F) \rightarrow \text{Gal}(K/F) \times \text{Gal}(L/F), \quad \sigma \mapsto (\sigma|_K, \sigma|_L)$$

is a group isomorphism.

Problem 2. Let K/F be a finite extension of fields and $f(x) \in F[x]$ an irreducible polynomial of degree n .

- (a) Prove that if n does not divide $[K : F]$ then $f(x)$ has no roots in K .

(b) Prove that if n is relatively prime to $[K : F]$ then $f(x)$ is irreducible over K .

Problem 3. Let F be a field, $f(x)$ a polynomial of degree n over F with splitting field E/F .

(a) Let K/F be a subextension of E/F . Prove that E/K is a splitting field of $f(x)$ considered as a polynomial over K .

(b) Prove that $[E : F]$ divides $n!$.

Hint. Use induction on n , and deal with the cases of f reducible or irreducible separately. At some point you will need the fact that $a!b!$ divides $(a + b)!$, which you should also prove (or have already proved in Math 71).

Problem 4. Let K/F be a Galois extension with Galois group isomorphic to $C_2 \times C_{12}$. How many subextensions $K/M/F$ are there satisfying:

(a) $[K : M] = 6$

(b) $[M : F] = 6$

(c) $\text{Gal}(K/M) \cong C_6$

(d) $\text{Gal}(M/F) \cong C_6$